```
#PROJECT ON PANDAS FUNCTIONS
```

```
import pandas as pd
```

```
import pandas as pd

data = {
    "Car_ID": [
        4832, 5127, 6941, 2756, 8394,
        6173, 7045, 5269, 3412, 9658,
        7884, 4391, 2106, 8753, 9902,
        6581, 7124, 3809, 5567, 8940
    ],
    "Car_Name": [
        "Maruti Swift", "Hyundai i20", "Honda City", "Tata Nexon", "Kia Seltos",
        "Mahindra XUV300", "Hyundai Creta", "Toyota Glanza", "Maruti Baleno", "Honda Amaze",
        "Renault Kiger", "Tata Altroz", "Nissan Magnite", "Volkswagen Polo", "Ford EcoSport",
        "Skoda Kushaq", "Hyundai Venue", "Kia Sonet", "Tata Punch", "Maruti Ciaz"
    ],
    "Mileage_kmpl": [
        22.0, 20.5, 18.4, 21.1, 19.8,
        20.0, 21.4, 22.3, 23.1, 19.2,
        20.2, 19.9, 20.5, 18.9, 17.8,
        19.5, 20.7, 21.0, 20.1, 20.8
    ],
    "Fuel_Type": [
        "Petrol", "Petrol", "Petrol", "Diesel", "Diesel",
        "Diesel", "Petrol", "Petrol", "Petrol", "Diesel",
        "Petrol", "Diesel", "Petrol", "Petrol", "Diesel",
        "Petrol", "Petrol", "Diesel", "Petrol", "Diesel"
    ]
}

df = pd.DataFrame(data)
```

```
df
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
0	4832	Maruti Swift	22.0	Petrol
1	5127	Hyundai i20	20.5	Petrol
2	6941	Honda City	18.4	Petrol
3	2756	Tata Nexon	21.1	Diesel
4	8394	Kia Seltos	19.8	Diesel
5	6173	Mahindra XUV300	20.0	Diesel
6	7045	Hyundai Creta	21.4	Petrol
7	5269	Toyota Glanza	22.3	Petrol
8	3412	Maruti Baleno	23.1	Petrol
9	9658	Honda Amaze	19.2	Diesel
10	7884	Renault Kiger	20.2	Petrol
11	4391	Tata Altroz	19.9	Diesel
12	2106	Nissan Magnite	20.5	Petrol
13	8753	Volkswagen Polo	18.9	Petrol
14	9902	Ford EcoSport	17.8	Diesel
15	6581	Skoda Kushaq	19.5	Petrol
16	7124	Hyundai Venue	20.7	Petrol
17	3809	Kia Sonet	21.0	Diesel
18	5567	Tata Punch	20.1	Petrol
19	8940	Maruti Ciaz	20.8	Diesel

Next steps: Generate code with df New interactive sheet

```
#Display the first 6 rows of the dataset to get an overview of the data
```

```
df.head(6)
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
0	4832	Maruti Swift	22.0	Petrol
1	5127	Hyundai i20	20.5	Petrol
2	6941	Honda City	18.4	Petrol
3	2756	Tata Nexon	21.1	Diesel
4	8394	Kia Seltos	19.8	Diesel
5	6173	Mahindra XUV300	20.0	Diesel

Next steps: Generate code with df New interactive sheet

```
#What happens if you run df.tail(10)? How many rows are shown?

df.tail(10)
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
10	7884	Renault Kiger	20.2	Petrol
11	4391	Tata Altroz	19.9	Diesel
12	2106	Nissan Magnite	20.5	Petrol
13	8753	Volkswagen Polo	18.9	Petrol
14	9902	Ford EcoSport	17.8	Diesel
15	6581	Skoda Kushaq	19.5	Petrol
16	7124	Hyundai Venue	20.7	Petrol
17	3809	Kia Sonet	21.0	Diesel
18	5567	Tata Punch	20.1	Petrol
19	8940	Maruti Ciaz	20.8	Diesel

```
#get a summary specifically for Mileage_kmpl

df["Mileage_kmpl"].describe()
```

	Mileage_kmpl
count	20.000000
mean	20.360000
std	1.291837
min	17.800000
25%	19.725000
50%	20.350000
75%	21.025000
max	23.100000

dtype: float64

```
#check the total number of rows and columns in the dataset

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 4 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Car_ID        20 non-null     int64
 1   Car_Name      20 non-null     object
 2   Mileage_kmpl  20 non-null     float64
 3   Fuel_Type     20 non-null     object
dtypes: float64(1), int64(1), object(2)
memory usage: 772.0+ bytes
```

```
#Selection & Indexing
#How do you select the Car_Name and Mileage_kmp columns from the DataFrame?
```

```
df[['Car_Name','Mileage_kmpl']]
```

	Car_Name	Mileage_kmpl
0	Maruti Swift	22.0
1	Hyundai i20	20.5
2	Honda City	18.4
3	Tata Nexon	21.1
4	Kia Seltos	19.8
5	Mahindra XUV300	20.0
6	Hyundai Creta	21.4
7	Toyota Glanza	22.3
8	Maruti Baleno	23.1
9	Honda Amaze	19.2
10	Renault Kiger	20.2
11	Tata Altroz	19.9
12	Nissan Magnite	20.5
13	Volkswagen Polo	18.9
14	Ford EcoSport	17.8
15	Skoda Kushaq	19.5
16	Hyundai Venue	20.7
17	Kia Sonet	21.0
18	Tata Punch	20.1
19	Maruti Ciaz	20.8

```
#How do you select the row for the student with Car_ID 2106?

df[df['Car_ID'] == 2106]
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
12	2106	Nissan Magnite	20.5	Petrol

```
#How do you select the first 3 rows and the first 2 columns?
df.iloc[:3, :2]
```

	Car_ID	Car_Name
0	4832	Maruti Swift
1	5127	Hyundai i20
2	6941	Honda City

```
#Filtering & Conditional Selection
#How do you find all Car_Name with a Mileage_kmpl greater than 19.9?

df[df['Mileage_kmpl'] >19.9]
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
0	4832	Maruti Swift	22.0	Petrol
1	5127	Hyundai i20	20.5	Petrol

```
#How do you filter Car_Names that use Petrol?

df[df['Fuel_Type'] == 'Petrol']
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
0	4832	Maruti Swift	22.0	Petrol
1	5127	Hyundai i20	20.5	Petrol
2	6941	Honda City	18.4	Petrol
6	7045	Hyundai Creta	21.4	Petrol
7	5269	Toyota Glanza	22.3	Petrol
8	3412	Maruti Baleno	23.1	Petrol
10	7884	Renault Kiger	20.2	Petrol
12	2106	Nissan Magnite	20.5	Petrol
13	8753	Volkswagen Polo	18.9	Petrol
15	6581	Skoda Kushaq	19.5	Petrol
16	7124	Hyundai Venue	20.7	Petrol
18	5567	Tata Punch	20.1	Petrol

```
#How do you select Car_Names from Hyundai with a Mileage_kmpl above 19.5 ?

df[(df['Car_Name'].str.contains('Hyundai')) & (df['Mileage_kmpl'] >19.5)]
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
1	5127	Hyundai i20	20.5	Petrol
6	7045	Hyundai Creta	21.4	Petrol
16	7124	Hyundai Venue	20.7	Petrol

```
#How do you find cars whose name starts with “M” or “N”?

df[df['Car_Name'].str.startswith(('M','N'))]
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
0	4832	Maruti Swift	22.0	Petrol
5	6173	Mahindra XUV300	20.0	Diesel
8	3412	Maruti Baleno	23.1	Petrol
12	2106	Nissan Magnite	20.5	Petrol
19	8940	Maruti Ciaz	20.8	Diesel

```
#Sorting
#How do you sort the DataFrame by Mileage_kmpl in descending order?

df.sort_values(by='Mileage_kmpl', ascending=False)
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
8	3412	Maruti Baleno	23.1	Petrol
7	5269	Toyota Glanza	22.3	Petrol
0	4832	Maruti Swift	22.0	Petrol
6	7045	Hyundai Creta	21.4	Petrol
3	2756	Tata Nexon	21.1	Diesel
17	3809	Kia Sonet	21.0	Diesel
19	8940	Maruti Ciaz	20.8	Diesel
16	7124	Hyundai Venue	20.7	Petrol
12	2106	Nissan Magnite	20.5	Petrol
1	5127	Hyundai i20	20.5	Petrol
10	7884	Renault Kiger	20.2	Petrol
18	5567	Tata Punch	20.1	Petrol
5	6173	Mahindra XUV300	20.0	Diesel
11	4391	Tata Altroz	19.9	Diesel
4	8394	Kia Seltos	19.8	Diesel
15	6581	Skoda Kushaq	19.5	Petrol
9	9658	Honda Amaze	19.2	Diesel
13	8753	Volkswagen Polo	18.9	Petrol
2	6941	Honda City	18.4	Petrol
14	9902	Ford EcoSport	17.8	Diesel

```
#How do you sort cars first by Car_Name and then by Fuel_Type?

df.sort_values(by=['Car_Name','Fuel_Type'])
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
14	9902	Ford EcoSport	17.8	Diesel
9	9658	Honda Amaze	19.2	Diesel
2	6941	Honda City	18.4	Petrol
6	7045	Hyundai Creta	21.4	Petrol
16	7124	Hyundai Venue	20.7	Petrol
1	5127	Hyundai i20	20.5	Petrol
4	8394	Kia Seltos	19.8	Diesel
17	3809	Kia Sonet	21.0	Diesel
5	6173	Mahindra XUV300	20.0	Diesel
8	3412	Maruti Baleno	23.1	Petrol
19	8940	Maruti Ciaz	20.8	Diesel
0	4832	Maruti Swift	22.0	Petrol
12	2106	Nissan Magnite	20.5	Petrol
10	7884	Renault Kiger	20.2	Petrol
15	6581	Skoda Kushaq	19.5	Petrol
11	4391	Tata Altroz	19.9	Diesel
3	2756	Tata Nexon	21.1	Diesel
18	5567	Tata Punch	20.1	Petrol
7	5269	Toyota Glanza	22.3	Petrol
13	8753	Volkswagen Polo	18.9	Petrol

```
#Aggregation & Grouping
#how do you calculate the average Mileage_kmpl for the entire dataset?

df['Mileage_kmpl'].mean()
```

```
np.float64(20.360000000000003)
```

```
 #How do you find the highest Mileage_kmpl for each Car_Name?

df.groupby('Car_Name')['Mileage_kmpl'].max()
```

Car_Name	Mileage_kmpl
Ford EcoSport	17.8
Honda Amaze	19.2
Honda City	18.4
Hyundai Creta	21.4
Hyundai Venue	20.7
Hyundai i20	20.5
Kia Seltos	19.8
Kia Sonet	21.0
Mahindra XUV300	20.0
Maruti Baleno	23.1
Maruti Ciaz	20.8
Maruti Swift	22.0
Nissan Magnite	20.5
Renault Kiger	20.2
Skoda Kushaq	19.5
Tata Altroz	19.9
Tata Nexon	21.1
Tata Punch	20.1
Toyota Glanza	22.3
Volkswagen Polo	18.9

dtype: float64

```
 #How do you count the number of cars from each Fuel_Type?

 df['Fuel_Type'].value_counts()
```

Fuel_Type	count
Petrol	12
Diesel	8

dtype: int64

```
 #how do you calculate the average Mileage_kmpl forcars grouped by Fuel_Type?

df.groupby('Fuel_Type')['Mileage_kmpl'].mean()
```

Fuel_Type	Mileage_kmpl
Diesel	19.950000
Petrol	20.633333

dtype: float64

```
#Adding & Modifying Columns
 #How do you increase every Mileage_kmpl by 0.1 for all cars?

df['Mileage_kmpl'] = df['Mileage_kmpl'] +0.1
df
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
0	4832	Maruti Swift	22.1	Petrol
1	5127	Hyundai i20	20.6	Petrol
2	6941	Honda City	18.5	Petrol
3	2756	Tata Nexon	21.2	Diesel
4	8394	Kia Seltos	19.9	Diesel
5	6173	Mahindra XUV300	20.1	Diesel
6	7045	Hyundai Creta	21.5	Petrol
7	5269	Toyota Glanza	22.4	Petrol
8	3412	Maruti Baleno	23.2	Petrol
9	9658	Honda Amaze	19.3	Diesel
10	7884	Renault Kiger	20.3	Petrol
11	4391	Tata Altroz	20.0	Diesel
12	2106	Nissan Magnite	20.6	Petrol
13	8753	Volkswagen Polo	19.0	Petrol
14	9902	Ford EcoSport	17.9	Diesel
15	6581	Skoda Kushaq	19.6	Petrol
16	7124	Hyundai Venue	20.8	Petrol
17	3809	Kia Sonet	21.1	Diesel
18	5567	Tata Punch	20.2	Petrol
19	8940	Maruti Ciaz	20.9	Diesel

```
#How do you create a new column Pass that is True if Mileage_kmpl >20.0 and False otherwise?

df['good_mileage'] = df['Mileage_kmpl'] >= 20.0
df
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type	good_mileage
0	4832	Maruti Swift	22.1	Petrol	True
1	5127	Hyundai i20	20.6	Petrol	True
2	6941	Honda City	18.5	Petrol	False
3	2756	Tata Nexon	21.2	Diesel	True
4	8394	Kia Seltos	19.9	Diesel	False
5	6173	Mahindra XUV300	20.1	Diesel	True
6	7045	Hyundai Creta	21.5	Petrol	True
7	5269	Toyota Glanza	22.4	Petrol	True
8	3412	Maruti Baleno	23.2	Petrol	True
9	9658	Honda Amaze	19.3	Diesel	False
10	7884	Renault Kiger	20.3	Petrol	True
11	4391	Tata Altroz	20.0	Diesel	True
12	2106	Nissan Magnite	20.6	Petrol	True
13	8753	Volkswagen Polo	19.0	Petrol	False
14	9902	Ford EcoSport	17.9	Diesel	False
15	6581	Skoda Kushaq	19.6	Petrol	False
16	7124	Hyundai Venue	20.8	Petrol	True
17	3809	Kia Sonet	21.1	Diesel	True
18	5567	Tata Punch	20.2	Petrol	True
19	8940	Maruti Ciaz	20.9	Diesel	True

Next steps: Generate code with df New interactive sheet

```
 #How do you create a new column Honor that shows "Yes" if GPA ≥ 3.7, otherwise "No"?

df['badMileage'] = df['Mileage_kmpl'].apply(lambda x: 'Yes' if x >= 20.0 else 'No')
df
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type	good_mileage	badMileage
0	4832	Maruti Swift	22.1	Petrol	True	Yes
1	5127	Hyundai i20	20.6	Petrol	True	Yes
2	6941	Honda City	18.5	Petrol	False	No
3	2756	Tata Nexon	21.2	Diesel	True	Yes
4	8394	Kia Seltos	19.9	Diesel	False	No
5	6173	Mahindra XUV300	20.1	Diesel	True	Yes
6	7045	Hyundai Creta	21.5	Petrol	True	Yes
7	5269	Toyota Glanza	22.4	Petrol	True	Yes
8	3412	Maruti Baleno	23.2	Petrol	True	Yes
9	9658	Honda Amaze	19.3	Diesel	False	No
10	7884	Renault Kiger	20.3	Petrol	True	Yes
11	4391	Tata Altroz	20.0	Diesel	True	Yes
12	2106	Nissan Magnite	20.6	Petrol	True	Yes
13	8753	Volkswagen Polo	19.0	Petrol	False	No
14	9902	Ford EcoSport	17.9	Diesel	False	No
15	6581	Skoda Kushaq	19.6	Petrol	False	No
16	7124	Hyundai Venue	20.8	Petrol	True	Yes
17	3809	Kia Sonet	21.1	Diesel	True	Yes
18	5567	Tata Punch	20.2	Petrol	True	Yes
19	8940	Maruti Ciaz	20.9	Diesel	True	Yes

Next steps: Generate code with df New interactive sheet

```
#String Operations
#How do you extract the first names from the Car_Name column?
df['First_Name'] = df['Car_Name'].str.split().str[0]
df
```

	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type	good_mileage	badMileage	First_Name
0	4832	Maruti Swift	22.1	Petrol	True	Yes	Maruti

```
import pandas as pd

data = {
    "Car_ID": [
        4832, 5127, 6941, 2756, 8394,
        6173, 7045, 5269, 3412, 9658,
        7884, 4391, 2106, 8753, 9902,
        6581, 7124, 3809, 5567, 8940
    ],
    "Car_Name": [
        "Maruti Swift", "Hyundai i20", "Honda City", "Tata Nexon", "Kia Seltos",
        "Mahindra XUV300", "Hyundai Creta", "Toyota Glanza", "Maruti Baleno", "Honda Amaze",
        "Renault Kiger", "Tata Altroz", "Nissan Magnite", "Volkswagen Polo", "Ford EcoSport",
        "Skoda Kushaq", "Hyundai Venue", "Kia Sonet", "Tata Punch", "Maruti Ciaz"
    ],
    "Mileage_kmpl": [
        22.0, 20.5, 18.4, 21.1, 19.8,
        20.0, 21.4, 22.3, 23.1, 19.2,
        20.2, 19.9, 20.5, 18.9, 17.8,
        19.5, 20.7, 21.0, 20.1, 20.8
    ],
    "Fuel_Type": [
        "Petrol", "Petrol", "Petrol", "Diesel", "Diesel",
        "Diesel", "Petrol", "Petrol", "Petrol", "Diesel",
        "Petrol", "Diesel", "Petrol", "Petrol", "Diesel",
        "Petrol", "Petrol", "Diesel", "Petrol", "Diesel"
    ]
}

df = pd.DataFrame(data)
```

```
df
```

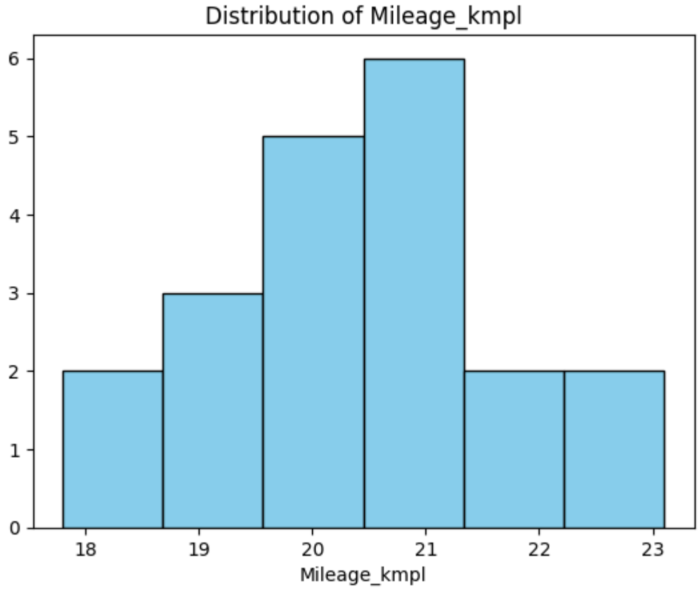
	Car_ID	Car_Name	Mileage_kmpl	Fuel_Type
0	4832	Maruti Swift	22.0	Petrol
1	5127	Hyundai i20	20.5	Petrol
2	6941	Honda City	18.4	Petrol
3	2756	Tata Nexon	21.1	Diesel
4	8394	Kia Seltos	19.8	Diesel
5	6173	Mahindra XUV300	20.0	Diesel
6	7045	Hyundai Creta	21.4	Petrol
7	5269	Toyota Glanza	22.3	Petrol
8	3412	Maruti Baleno	23.1	Petrol
9	9658	Honda Amaze	19.2	Diesel
10	7884	Renault Kiger	20.2	Petrol
11	4391	Tata Altroz	19.9	Diesel
12	2106	Nissan Magnite	20.5	Petrol
13	8753	Volkswagen Polo	18.9	Petrol
14	9902	Ford EcoSport	17.8	Diesel
15	6581	Skoda Kushaq	19.5	Petrol
16	7124	Hyundai Venue	20.7	Petrol
17	3809	Kia Sonet	21.0	Diesel
18	5567	Tata Punch	20.1	Petrol
19	8940	Maruti Ciaz	20.8	Diesel

Next steps: Generate code with df New interactive sheet

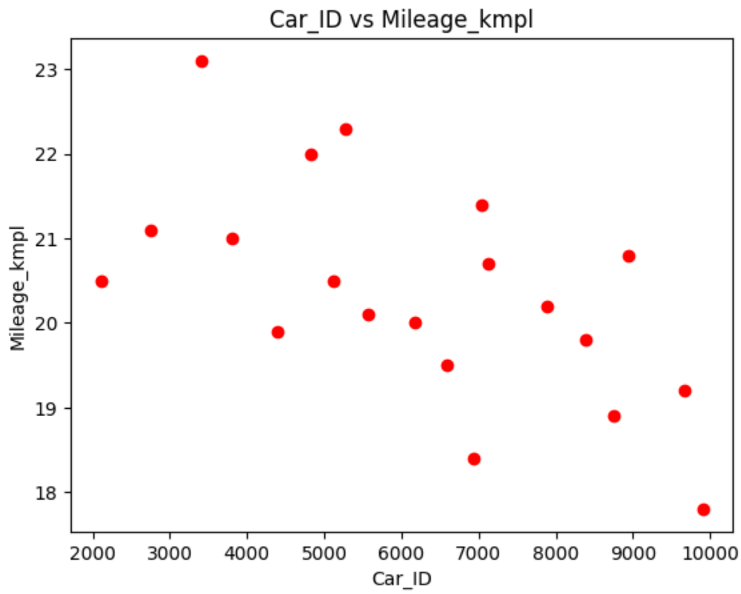
```
 #GRAPHS
 #HISTOGRAM GRAPH
 #Plot a histogram of the cars’ Mileage_kmpl.

import matplotlib.pyplot as plt
```

```
plt.hist(df['Mileage_kmpl'], bins=6, color='skyblue', edgecolor='black')
plt.title('Distribution of Mileage_kmpl')
plt.xlabel('Mileage_kmpl')
```

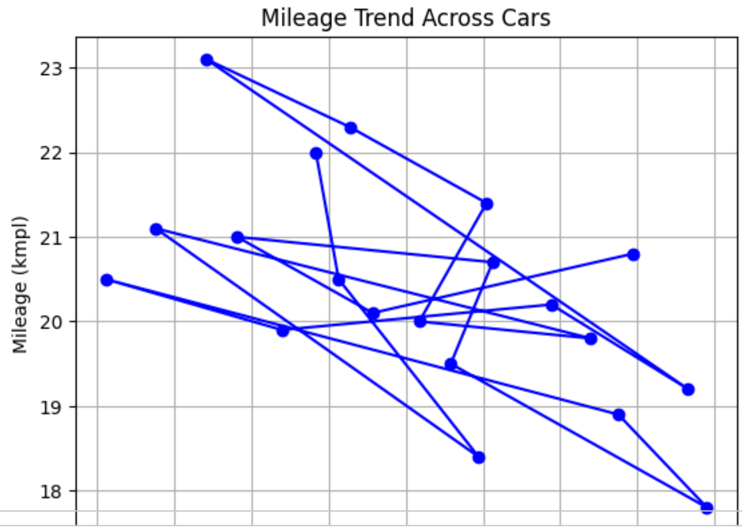
```
Text(0.5, 0, 'Mileage_kmpl')
```




```
 #SCATTER GRAPH
 #Plot Car_Id vs Mileage_kmpl to see individual performance trends.

import matplotlib.pyplot as plt

plt.scatter(df['Car_ID'], df['Mileage_kmpl'], color='red')
plt.title('Car_ID vs Mileage_kmpl')
plt.xlabel('Car_ID')
plt.ylabel('Mileage_kmpl')
plt.show()
```




```
 # LINE GRAPH
# Plot Mileage_kmpl against Car_ID as a line plot to see mileage trends across cars.
import matplotlib.pyplot as plt

plt.plot(df['Car_ID'], df['Mileage_kmpl'], marker='o', color='blue', linestyle='-')
plt.title('Mileage Trend Across Cars')
plt.xlabel('Car_ID')
plt.ylabel('Mileage (kmpl)')
plt.grid(True)
plt.show()
```




```
# BAR GRAPH
# Calculate average Mileage_kmpl per Car_Name
avg_mileage = df.groupby('Car_Name')['Mileage_kmpl'].mean().sort_values(ascending=False)

# Plot bar graph
avg_mileage.plot(kind='bar', color='skyblue', edgecolor='black')
plt.title('Average Mileage per Car Name')
plt.xlabel('Car_Name')
plt.ylabel('Average Mileage (kmpl)')
plt.xticks(rotation=45, ha='right')
plt.show()
```